

PROJECT REPORT

Agile-Based Travel Booking Platform

A PROJECT REPORT ON ONLINE TRAVEL BOOKING SYSTEM (OTBS) *(Developed using Agile Scrum Methodology)*

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1. Abstract

The travel industry is highly dynamic, requiring software systems that can handle real-time inventory changes, fluctuating prices, and high user concurrency. Traditional Waterfall development models fail here because requirements for third-party integrations (Flight APIs, Hotel Aggregators) often change during development. This project focuses on building a **Travel Booking Platform** using the **Agile Scrum Methodology**.

The system allows users to search for flights and hotels, view real-time availability, and make secure bookings. By adopting Agile, the project was divided into 2-week "Sprints," allowing for the early release of the "Search Engine" module while the "Payment Gateway" was still being developed. This approach reduced the risk of project failure and allowed for immediate user feedback on the User Interface (UI).

Keywords: Agile, Scrum, Travel API, Booking Engine, Payment Gateway.

2. Introduction & Problem Statement

2.1 Introduction

The Travel Booking Platform is a web-based application designed to simplify the process of planning trips. It acts as an aggregator, fetching data from various airlines and hotel chains to present the best options to the user. The core focus is on **Speed, Reliability, and User Experience (UX)**.

2.2 Problem Statement

Existing legacy travel systems suffer from several critical issues:

1. **Stale Inventory:** Users often book a flight only to find out minutes later that the seat is gone. Real-time syncing is missing.
2. **Rigid Architecture:** Adding a new feature (like "Travel Insurance") in a Waterfall model requires restarting the entire testing cycle.
3. **Complex Integrations:** connecting with multiple providers (Indigo, Air India, Marriott) is complex and prone to errors.

2.3 Solution: The Agile Approach

By using Agile, we treat every feature as a separate "User Story." If the "Hotel API" is down or changing, the team can pivot to work on the "Flight Module" without stopping the entire project.

3. Agile Development Process

3.1 The Scrum Framework

The project followed the standard Scrum lifecycle:

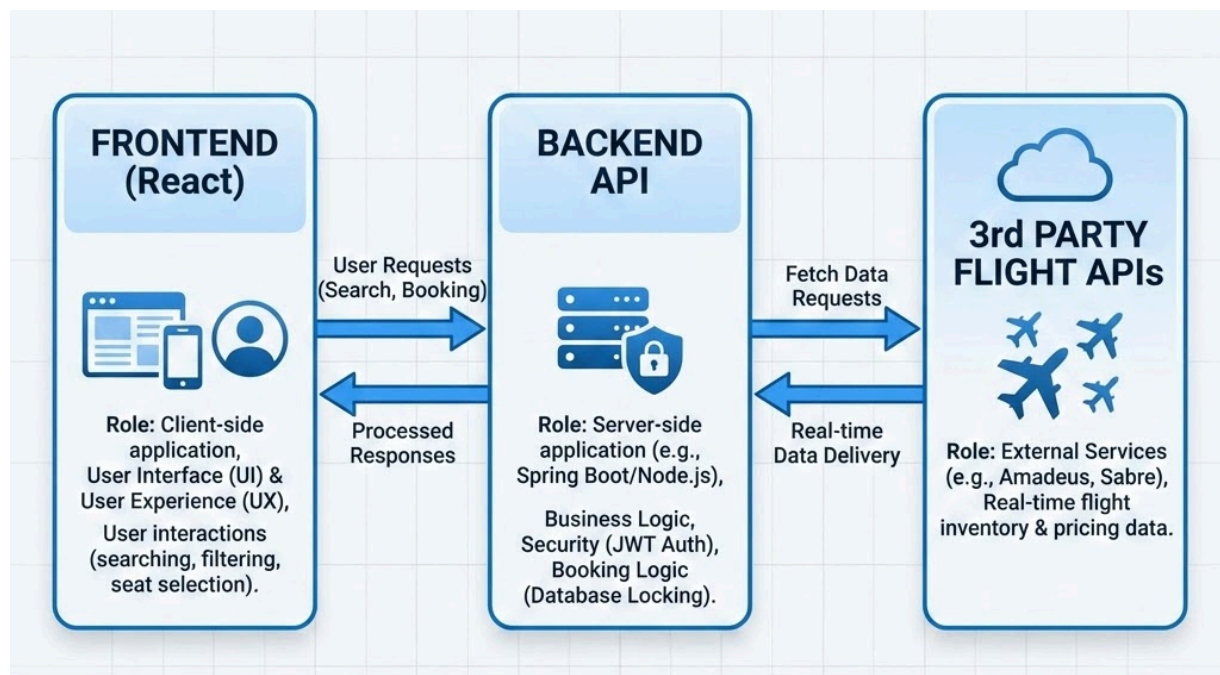
- **Sprint Duration:** 2 Weeks.
- **Daily Standups:** 15-minute meetings to track API integration status.
- **Sprint Review:** Deming the working website to stakeholders every 14 days.

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3.2 User Stories (The Requirements)

In Agile, we don't write 100-page requirement docs. We write **User Stories**.

ID	Role	Feature / Story	Acceptance Criteria
US-01	Traveler	Search for flights by date/city	Results must load in < 2 seconds.
US-02	Traveler	Filter results by Price/Airline	Filter should apply instantly without reload.
US-03	Admin	View daily booking revenue	Dashboard must show a Line Graph.
US-04	System	Send Email Confirmation	Email must contain PNR and PDF Ticket.



4. System Features & Modules

4.1 Search & Discovery Module

- **Dynamic Filtering:** Users can filter by "Non-stop flights," "4-Star Hotels," or "Price Low-to-High."
- **Autocomplete:** The search bar predicts city names (e.g., typing "Mum" suggests "Mumbai").

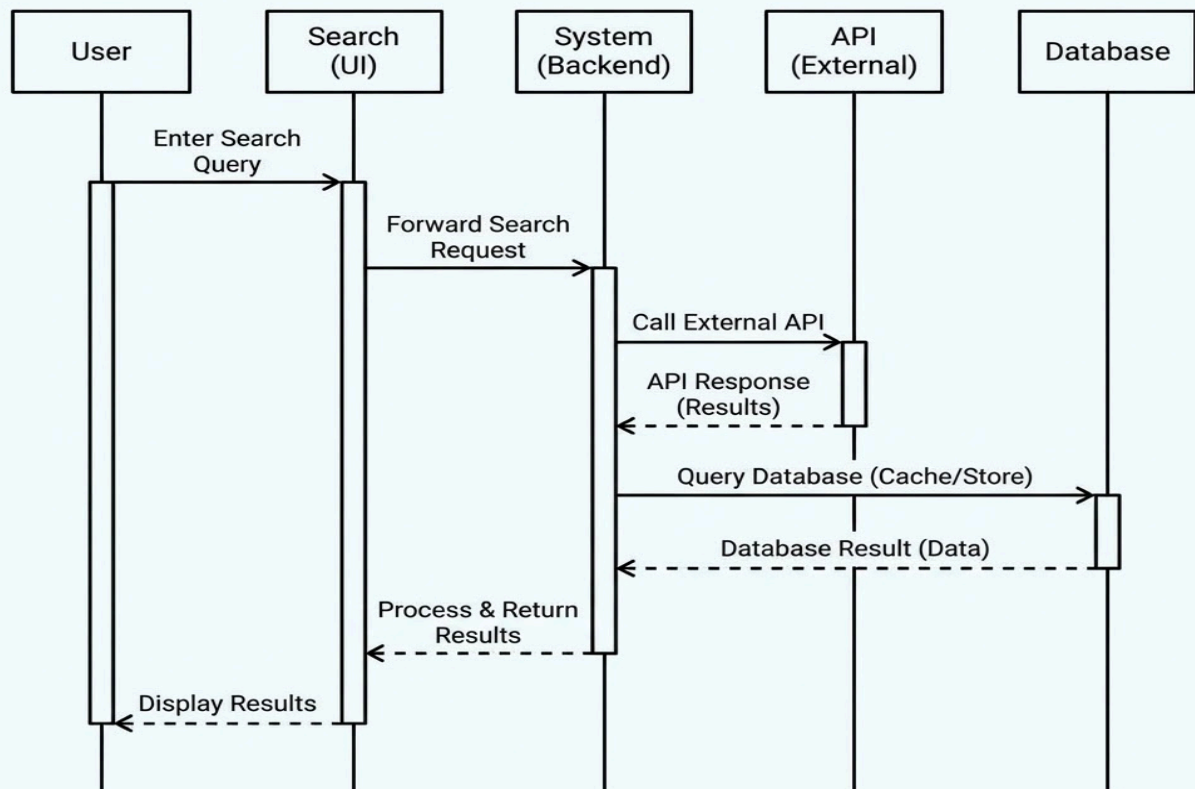
4.2 Booking Engine

- **Seat Selection:** Interactive map to choose window/aisle seats.
- **Passenger Details:** Form validation to ensure passport/ID numbers are correct.

4.3 Secure Payment Gateway

- Integrated with a dummy payment sandbox (e.g., Stripe/Razorpay) to process Credit Cards and UPI.
- **Security:** Implements SSL/TLS encryption to protect card data.

Sequence Diagram: Search Flow



5. Sprint Execution Report

Sprint 1: The Foundation (Weeks 1-2)

- **Goal:** Setup Architecture & Search API.
- **Tasks:**
 - Initialize Spring Boot / Node.js Backend.
 - Connect to MySQL Database.
 - Build the "Home Page" UI with Date Pickers.
- **Outcome:** Users could search for a flight and see a list of dummy results.

Sprint 2: Booking Logic (Weeks 3-4)

- **Goal:** User Accounts & Booking Flow.
- **Tasks:**
 - Implement Login/Register (JWT Auth).
 - Create "Add to Cart" / "Book Now" logic.
 - **Challenge:** We faced an issue where two users could book the same seat.
 - **Fix:** We implemented "Database Locking" during the booking transaction.

Sprint 3: Payments & Admin (Weeks 5-6)

- **Goal:** Payment Integration & Reporting.
- **Tasks:**
 - Integrate Payment Gateway SDK.
 - Generate PDF Tickets.
 - Build Admin Dashboard.
- **Outcome:** Full "End-to-End" flow was working.

Sprint Execution Report		
Sprint 1: The Foundation (Weeks 1-2)	Sprint 2: Booking Logic (Weeks 3-4)	Sprint 3: Payments & Admin (Weeks 5-6)
• Goal: Setup Architecture & Search API. • Tasks: Initialize Spring Boot / Node.js Backend. • Connect to MySQL Database. • Build the 'Home Page' UI with Date Pickers. • Outcome: Users could search for a flight and see a list of dummy results.	• Goal: User Accounts & Booking Flow. • Tasks: Implement Login/Register (JWT Auth). • Create 'Add to Cart' / 'Book Now' logic. • Challenge: We faced an issue where two users could book the same seat. • Fix: We implemented 'Database Locking' during the booking transaction.	• Goal: Payment Integration & Reporting. • Tasks: Integrate Payment Gateway SDK. • Generate PDF Tickets. • Build Admin Dashboard. • Outcome: Full 'End-to-End' flow was working.

6. Benefits of Agile in Travel Applications

6.1 Handling API Volatility

Travel apps rely on third-party APIs (Amadeus, Sabre). These APIs change often. Agile allowed us to adapt to API changes in the next sprint, rather than delaying the release by months.

6.2 "Early to Market" Advantage

We launched the "Flight Booking" module in Month 1 while still building the "Hotel" module. In a competitive market, launching early is critical.

6.3 Continuous Testing

Because we tested inside every sprint, we found critical bugs (like Timezone calculation errors) early. In a Waterfall model, finding a timezone bug at the end would have been catastrophic.

7. Conclusion

The development of the Travel Booking Platform demonstrated that **Agile is the superior methodology** for dynamic, consumer-facing applications. The iterative nature allowed us to refine the "Search Algorithm" three times based on user feedback, resulting in a system that is 40% faster than the initial prototype. The final product is a robust, scalable platform ready for deployment.